

ARMAN SINGH GREWAL

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EXPERIENCE

Software Engineer

Xperi (TiVo) - Bengaluru, KA

Tech Stack: **Python, FastAPI, DuckDB, Polars, Docker, Elasticsearch**

May 2025 - present

- Automated feed profiling for **40+** metadata providers across **10+** formats by building a pre-ingest DQ platform with **Polars** and **DuckDB**, eliminating manual validation before central pipeline ingestion.
- Led migration of 50+ data pipeline scripts from **CentOS** to **RockyOS**, including Python 3.5→3.9 upgrades, **Elasticsearch** SSL auth, and **dockerization** of 6+ components; ensuring zero disruption to metadata pipelines serving **40+** providers.
- Resolved **3** critical security vulnerabilities in the internal KGVt platform - including unrestricted API access, unauthenticated search endpoints enabling data scraping, and exposed system internals, by implementing **Django**-based session authentication and token-level API access controls.
- Migrated 40+ job notification pipelines from Slack to Microsoft Teams by building a Python wrapper over the **PowerAutomate** API with AdaptiveCards, replacing Slack SDK dependency across the entire notification infrastructure.

Software Engineer Trainee (R&D)

Informatica - Bengaluru, KA

Tech Stack: **Python, FastAPI, Apache Airflow, React, Next**

Aug 2024 - May 2025

- Built a cross-cloud Resource Utilization Dashboard (AWS, GCP, Azure, OCI) in **Next.js** and **Highcharts**, with interactive region → cluster → node → service drill-down; part of an initiative that drove **\$9M** in YTD cost savings.
- Implemented lazy-loaded tree views and **SWR polling** (15-min intervals) matched to **Airflow**'s ingestion cadence, avoiding upfront fetch of 600 nodes and services across the 4 cloud providers.

Software Engineer Intern

Xperi (TiVo) - Bengaluru, KA

Tech Stack: **Python, Bash, SQL, NoSQL**

Aug 2023 - Jun 2024

- Built and enhanced metadata collection, enrichment, and processing pipelines for movies and episodes, improving throughput and reliability of delivery to **40+** downstream partner systems.
- Cut data sync latency by **94%** (72 hrs → 4 hrs) by redesigning the Python synchronization framework between **Elasticsearch** and **MSSQL**, removing bottlenecks in the metadata consistency layer.
- Implemented a monitoring pipeline using **Telegraf** to collect **Kafka** consumer lag metrics from **Burrow** and visualize them in **Splunk** via interactive dashboards.

PROJECTS

PaathGuide (Gurbani Audio Search & Recognition) 🌐 🔗

May 2025

Tech Stack: **Next.js 14, React, Tailwind CSS, SQLite, Groq SDK (Whisper ASR)**

- Built a **Next.js** PWA to capture live audio and transcribe recitation via **OpenAI Whisper**, using confidence filtering (*avg_logprob, no_speech_prob*) to eliminate noise hallucinations.
- Engineered a 4-stage phonetic normalization pipeline in JavaScript to strip vowel diacritics and collapse aspirated consonant pairs, resolving transcription mismatches against a canonical SQLite database.
- Designed a Pauree-level **TF-IDF** search engine with length normalization ($\sqrt{|vocab|}$) and prefix-match fallback ($0.5 \times \text{weight}$) to score and rank scripture verses in real-time under ambient Gurudwara acoustic environments.

SKILLS

Languages	Python, Shell scripting, Java, C++, C, SQL
Web Technologies	Node, Express, Vue, React, Next, Flask, Django, Spring Boot
Databases	MySQL, MSSQL, PostgreSQL, Redis, ElasticSearch, Neo4j, MongoDB
Machine Learning	Scikit-learn, Tensorflow, Pytorch, Keras, Splunk, Excel, MatLab
DevOps	Docker, Jenkins, RunDeck, Kubernetes

ACHIEVEMENTS AND CERTIFICATIONS

- Neo4j** Certified Professional
- Specialization in **Natural Language Processing**
- Level: **Contributor** on **Kaggle**
- Redis** Certified Developer
- RH-124** and **RH-134** from **RedHat**
- DBMS-1** and **DBMS-2** from **Infosys**

EDUCATION

BTech Computer Science and Engineering (Hons.)

Lovely Professional University, Phagwara, Punjab

2020 - 2024

CGPA 8.84